MANAV RACHNA INTERNATIONAL INSTITUTE OF RESEARCH & STUDIES
SCHOOL OF COMPUTER APPLICATIONS

PRACTICAL FILE
Data Structure and Algorithm
Academic Session: 2026–2027

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1. Deque Operations

main.py

Visualize Run ►

```
deque = []

while True:
    print("\n1. Insert at front")
    print("2. Insert at rear")
    print("3. Delete from front")
    print("4. Delete from rear")
    print("5. Display")
    print("6. Exit")

    choice = input("Choose an option: ")

    if choice == "1":
        value = int(input("Enter value: "))
        deque.insert(0, value)
        print("Value added at front")

    elif choice == "2":
        value = int(input("Enter value: "))
        deque.append(value)
        print("Value added at rear")

    elif choice == "3":
        if len(deque) == 0:
            print("Deque is empty")
        else:
            print("Deleted:", deque.pop(0))

    elif choice == "4":
        if len(deque) == 0:
            print("Deque is empty")
        else:
            print("Deleted:", deque.pop())

    elif choice == "5":
        print("Deque:", deque)

    elif choice == "6":
        print("Program ended")
        break

    else:
        print("Invalid choice")
```

Output

```
1. Insert at front
2. Insert at rear
3. Delete from front
4. Delete from rear
5. Display
6. Exit
Choose an option: 2
Enter value: 10
Value added at rear

Choose an option: 5
Deque: [10]
```

2. Queue Operations

main.py Visualize Run ►

```
queue = []

while True:
    print("\n1. Insert")
    print("2. Delete")
    print("3. Display")
    print("4. Exit")

    choice = input("Choose an option: ")

    if choice == "1":
        value = int(input("Enter value: "))
        queue.append(value)
        print("Value added")

    elif choice == "2":
        if len(queue) == 0:
            print("Queue is empty")
        else:
            print("Deleted:", queue.pop(0))

    elif choice == "3":
        print("Queue:", queue)

    elif choice == "4":
        print("Program ended")
        break

    else:
        print("Invalid choice")
```

Output

```
1. Insert
2. Delete
3. Display
4. Exit
Choose an option: 1
Enter value: 25
Value added

Choose an option: 3
Queue: [25]
```

3. Stack Operations

main.py

Visualize Run ►

```
stack = []

while True:
    print("\n1. Push")
    print("2. Pop")
    print("3. Show top")
    print("4. Display")
    print("5. Exit")

    choice = input("Choose an option: ")

    if choice == "1":
        value = int(input("Enter value: "))
        stack.append(value)
        print("Value added")

    elif choice == "2":
        if len(stack) == 0:
            print("Stack is empty")
        else:
            print("Removed:", stack.pop())

    elif choice == "3":
        if len(stack) == 0:
            print("Stack is empty")
        else:
            print("Top value:", stack[-1])

    elif choice == "4":
        print("Stack:", stack[::-1])

    elif choice == "5":
        print("Program ended")
        break

    else:
        print("Invalid choice")
```

Output

```
--- Stack Operations ---
1. Push
2. Pop
3. Show top
4. Display
5. Exit
Choose an option: 1
Enter value: 50
Value added

Choose an option: 4
Stack: [50]
```

4. Multiplication Table

main.py

Visualize Run ►

```
num = int(input("Enter a number: "))

for i in range(1, 11):
    result = num * i
    print(num, "x", i, "=", result)
```

Output

```
Enter a number: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

5. Factorial Calculation

main.py

Visualize Run ►

```
num = int(input("Enter a number: "))

if num < 0:
    print("Factorial is not possible")
else:
    answer = 1
    for i in range(1, num + 1):
        answer = answer * i
    print("Factorial =", answer)
```

Output

```
Enter a number: 5
Factorial = 120
```